

SilphNet

Technical Architecture & Feature Specification for Gen1Recomp MMO Framework

Target Engine: Gen1Recomp (LÖVE2D / Lua)

Author: Independent Software Engineer

Backend Stack: Go / Node.js + SQLite

Deployment: \$5/mo Linux VPS (Single Box)

1. Executive Summary & Core Vision

SilphNet is an open-source, auto-connecting multiplayer framework engineered specifically for **Gen1Recomp**. While existing community mods (such as *Gen1Online*) rely on fragmented peer-to-peer hosting, manual IP address entry, and complex in-game PC terminal menus, **SilphNet** delivers a seamless, zero-configuration "log in and play" overworld MMO experience.

By combining a lightweight **Lua client mod** running in a non-blocking background thread with a high-performance **Go / Node.js spatial server** hosted on a single \$5/month Linux VPS, SilphNet renders other trainers across Kanto in real-time. It integrates a native Friends List overlay, anti-cheat validated PvP battling, and cloud trading—all without requiring end users to manage ports or network settings.

2. System Architecture Overview

SilphNet uses a client-server architecture where the client acts as a thin renderer/input collector, while the server maintains world state, spatial partitioning, and security validation.

Binary Packet Payload Layout

To minimize bandwidth and CPU overhead, positional state is transmitted as a raw binary buffer (~10 bytes) at a **10 Hz tick rate** (10 updates/second):

Field Offset	Field Name	Data Type	Description
0x00 - 0x01	player_id	uint16 (2B)	Unique session ID assigned upon login
0x02 - 0x03	map_id	uint16 (2B)	Active map location ID
0x04	x_tile	uint8 (1B)	X-coordinate on tile grid (0–255)
0x05	y_tile	uint8 (1B)	Y-coordinate on tile grid (0–255)
0x06	facing_dir	uint8 (1B)	Facing direction (0=Down, 1=Up, 2=Left, 3=Right)
0x07	sprite_id	uint8 (1B)	Selected overworld trainer avatar
0x08	anim_state	uint8 (1B)	Movement animation frame index

4. Client UI & Social Experience

- **Authentication & Profiles:** JWT-based persistent logins stored locally on the client device.
- **Start Menu Social Overlay:** Custom Lua canvas drawing hooks into Gen1Recomp's pause menu to render:
 - **Friends List:** Shows online status, active route/town location, and lead party Pokémon.
 - **Direct Overworld Interaction:** Facing an online player and pressing **A** opens a contextual modal: **Battle Challenge**, **Trade Request**, or **Profile View**.
- **Lore Integration:** Named **SilphNet** after Silph Co., the primary tech corporation spanning both Kanto and Johto (powers Magnet Train, Pokégear, and teleporters).

5. Anti-Cheat & Competitive Integrity Engine

Since local save files on Android/PC can be edited with memory tools, the server acts as an authoritative referee before allowing competitive PvP battles or trades.

Gen 1 HP Calculation Formula:

$$HP = \lfloor ((Base + DV) \times 2 + \lfloor \sqrt{(StatEXP) / 4} \rfloor) \times Level / 100 \rfloor + Level + 10$$

Gen 1 Stat Calculation Formula (Attack, Defense, Speed, Special):

$$Stat = \lfloor ((Base + DV) \times 2 + \lfloor \sqrt{(StatEXP) / 4} \rfloor) \times Level / 100 \rfloor + 5$$

- **Strict Bounds Enforcement:** DVs bounded to [0, 15]; Stat EXP bounded to [0, 65535]. Any mismatch results in instant disconnection.
- **Movepool Whitelisting:** Movesets are validated against a canonical Gen 1 JSON lookup table.

6. Infrastructure & Deployment Specification

- **Recommended VPS Spec:** 1–2 vCPUs, 1–2 GB RAM, 25 GB NVMe SSD, 1–2 TB Monthly Bandwidth (~\$5/month on Hetzner or DigitalOcean).
- **Estimated Capacity:** 1,000 to 5,000+ concurrent players on a single \$5 VPS node.

- **Technology Stack:** Caddy (Auto HTTPS/WSS SSL), Go binary under systemd, SQLite database.

7. Legal & Monetization Framework

- **Non-Profit Fan Project:** 100% free and open-source. No paid items, microtransactions, cosmetics, or subscription paywalls.
- **Transparent Hardware Expense Funding:** Server costs covered via a public tip jar (Ko-fi or Open Collective) capped at the monthly VPS bill (\$5/month). Zero in-game perks given for donations to avoid Nintendo copyright/commercialization enforcement.

8. Concrete Prompts for Claude Code Scaffold

Prompt 1: Backend Server (Go + WebSockets + SQLite)

Create a high-performance, single-binary Go backend using Gorilla WebSockets and SQLite. Implement JWT token authentication (`/api/register`, `/api/login`), a `Friends` table, and a 10 Hz spatial WebSocket handler (`/ws`). Group connected clients by `map_id` and broadcast binary positional packets (`player_id`, `map_id`, `x`, `y`, `facing_dir`, `sprite_id`) only to clients in the same map.

Prompt 2: Anti-Cheat & Stat Calculator Module

Write a standalone Go package that validates Gen 1 Pokémon party data. Implement exact Gen 1 HP and Stat calculation formulas using Base Stats, DVs (0-15), Stat EXP (0-65535), and Level. Compare calculated stats against incoming client stats and include a movepool validator function using an embedded JSON map of legal Gen 1 move IDs per Pokémon.

Prompt 3: Lua Client Mod for Gen1Recomp

Write a LOVE2D Lua client mod module for Gen1Recomp. Initialize a non-blocking WebSocket connection inside a secondary `love.thread`. Send a 10-byte positional packet whenever the player moves tiles, parse incoming neighbor positional updates, and provide a global table `SilphNet.NearbyPlayers` so the main render loop can draw remote player sprites smoothly.

Prompt 4: Caddy & Systemd Deployment Scripts

Create a Dockerfile, Caddyfile (handling reverse proxying for `wss://silphnet.yourdomain.com` with auto-SSL), and a Systemd service file (`silphnet.service`) to deploy this Go binary and SQLite database on an Ubuntu 24.04 Linux VPS.